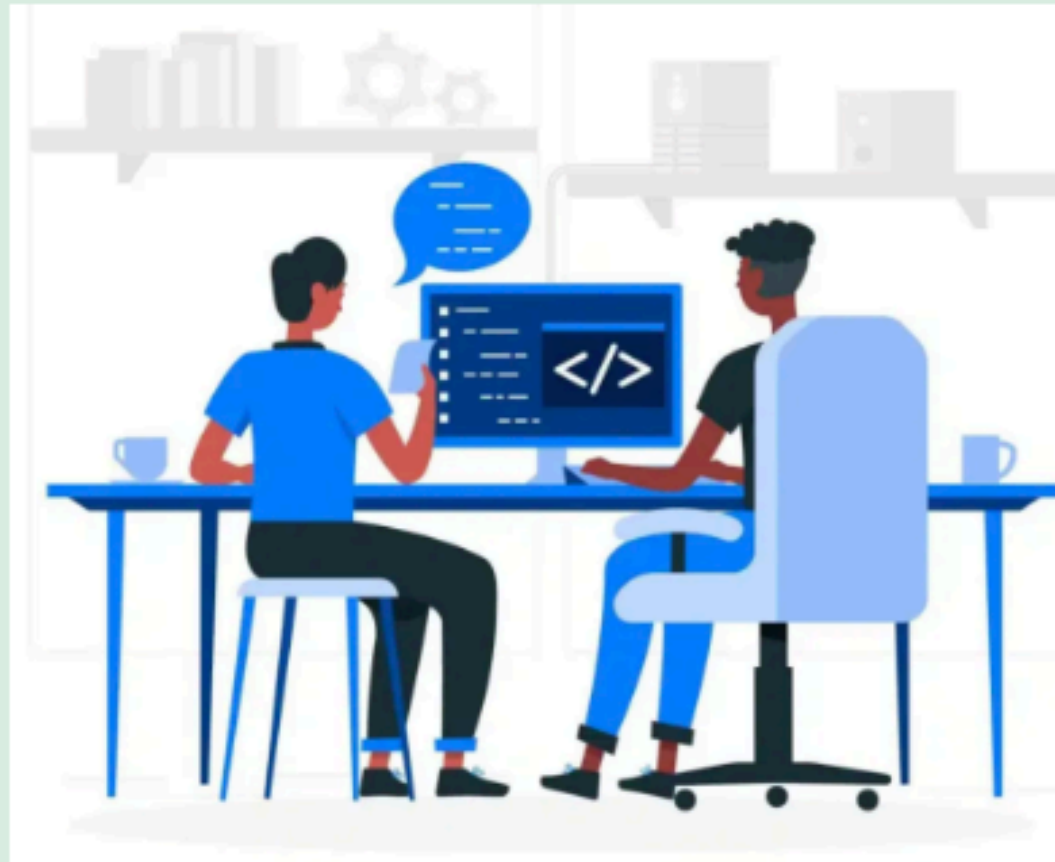


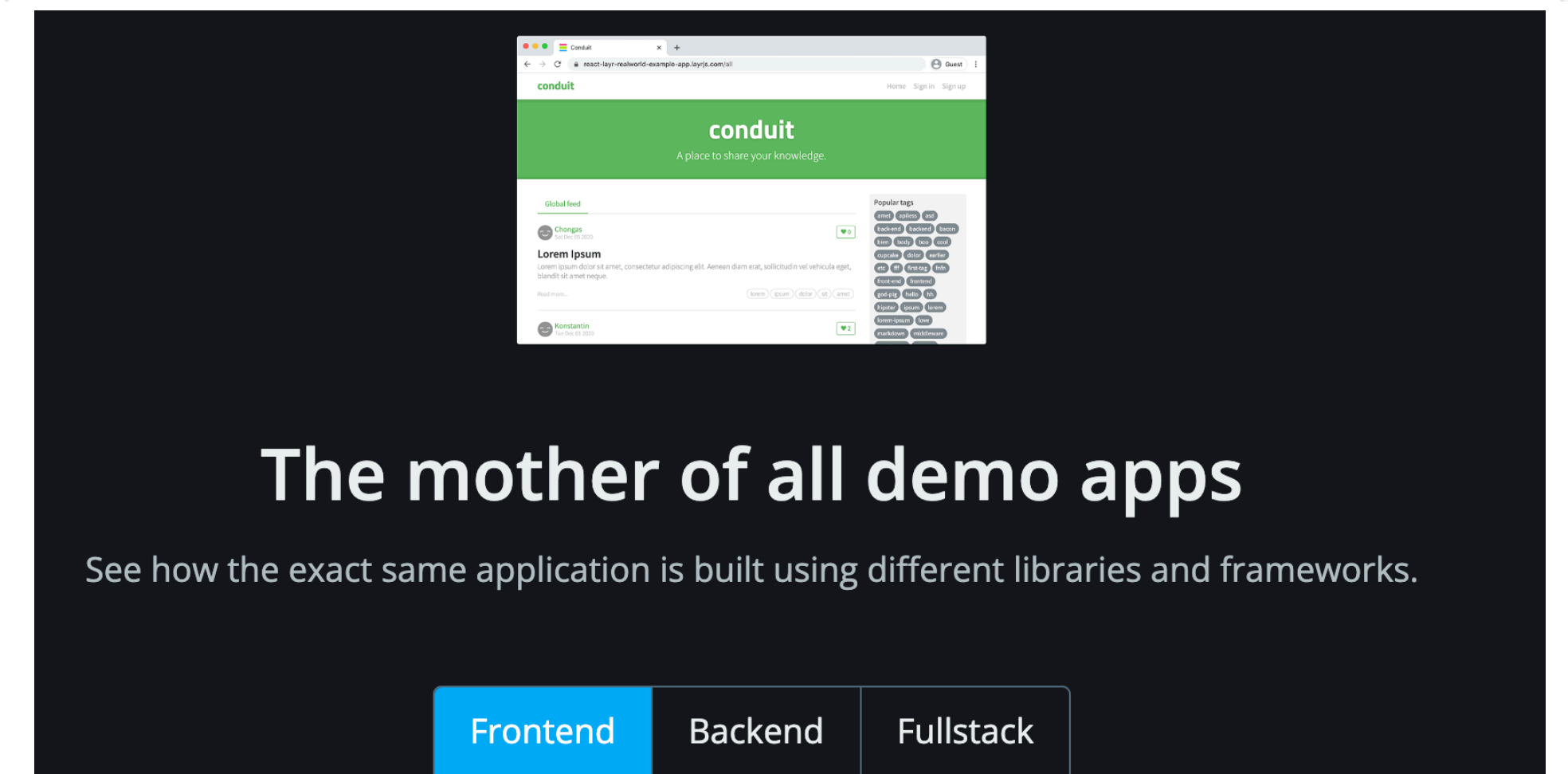
13: Assignment



Concept, detail and
grading rubric

Assignment Motivation & Philosophy

- Assignments present multiple opportunities:
 - Crystallise knowledge on module
 - Opportunity for creativity & innovation
 - Potentially important portfolio contribution
- It is not over when submitted, presented and graded
 - Ongoing maintenance & enhancements
 - Periodically rebuild using new technology stack



The mother of all demo apps

See how the exact same application is built using different libraries and frameworks.

Frontend

Backend

Fullstack

<https://github.com/gothinkster/realworld>
<https://codebase.show/projects/realworld>

Agenda

Assignment Briefing

Assignment Versions (choice)

Grading Rubric(s)

Assignment Briefing

Point of Interest

A point of interest, or POI, is a specific point location that someone may find useful or interesting.



“A “point of interest” (POI) is a location for which information is available. A POI can be as simple as a set of coordinates, a name, and a unique identifier, or more complex such as a three-dimensional model of a building with names in multiple languages, information about opening and closing hours, and a civic address. POI data has many applications, including augmented reality browsers, location-based social networking games, geocaching, mapping and navigation systems.”

<http://www.opengeospatial.org/projects/groups/poiswg>

https://en.wikipedia.org/wiki/Point_of_interest

https://en.wikipedia.org/wiki/Point_of_interest

PlaceMark Examples

- Landscape feature
- National monument
- Walking Trail
- Bridge
- Tree
- Venue
- Ringfort
- Dolmen
- River
- Bog
- Island
- Forest
- Beach
- Museum
- Climb
- Cycleway
- Birdwatching Locations
- Diving

Product Feature Sets

POI Sharing /
Community Features

Community

- You are free to innovate an the type, range and variety of features

Richer Media
Experience + User
Management

Enhanced

- These categories are to stimulate your brainstorming

Essential Data
Structure & Support
Service

Core

- Some ideas on the next slides...

Information Model

- Placemark
 - Name
 - Category
 - Description
 - Analytics
- User
 - Basic user details

Features

- Sign up / log in to service
- Create, Read, Update, Delete
- Query
- Group into Categories

Core

Information Model

- Placemark
 - Images
 - Location
 - Weather
- User
 - Types

Features

- Placemark images
- Placemark Map Location
- Admin
 - Admin user
 - Admin dashboard

Enhanced

Information Model

- Placemark
 - Creator / Editor
 - Rating
 - Reviews
 - Discussion
- User
 - Interaction information

Features

- Share
- Review & Rate
- Notice Board
- Live Updates
- Analytics

Community

Assignment Versions

PlacemarkCore



PlacemarkHAPI



OR

PlacemarkSveltekit





Implement POI model

Support the following User Management Features:

- Sign up / Login in / Delete Account
- Simple User Admin Dashboard

Support the following POI Characteristics:

- Name, Description, Category, Location, Image

Support the following app features:

- Create, Read, Update & Delete POIs
- Organise PlaceMarks into categories

API

- Endpoints + tests
- Swagger Documentation
- JWT Security

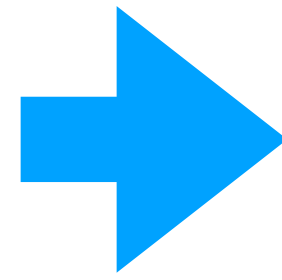
PlacemarkCore



Grading Levels

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
Level 1	Signup / Login Cookie Authentication/	Name, Description, Location (lat,lng)	Basic API, Core unit Tests	JSON	Locahost	git + commit History, Structured readme
Level 2	Basic Admin Account List/Remove Users	Categories + images	Open API (Swagger) + tests	Mongo	Cloud / Render	Tagged releases, Feature Branches

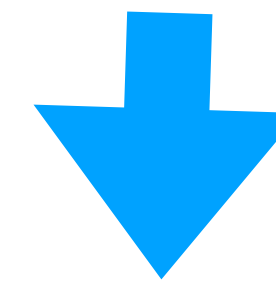
Having
completed this



PlacemarkCore



Then proceed
with one of



PlacemarkHAPI



Option A

?

OR

PlacemarkSveltekit



Option B

PlacemarkHAPI



- Choose

PlacemarkHAPI

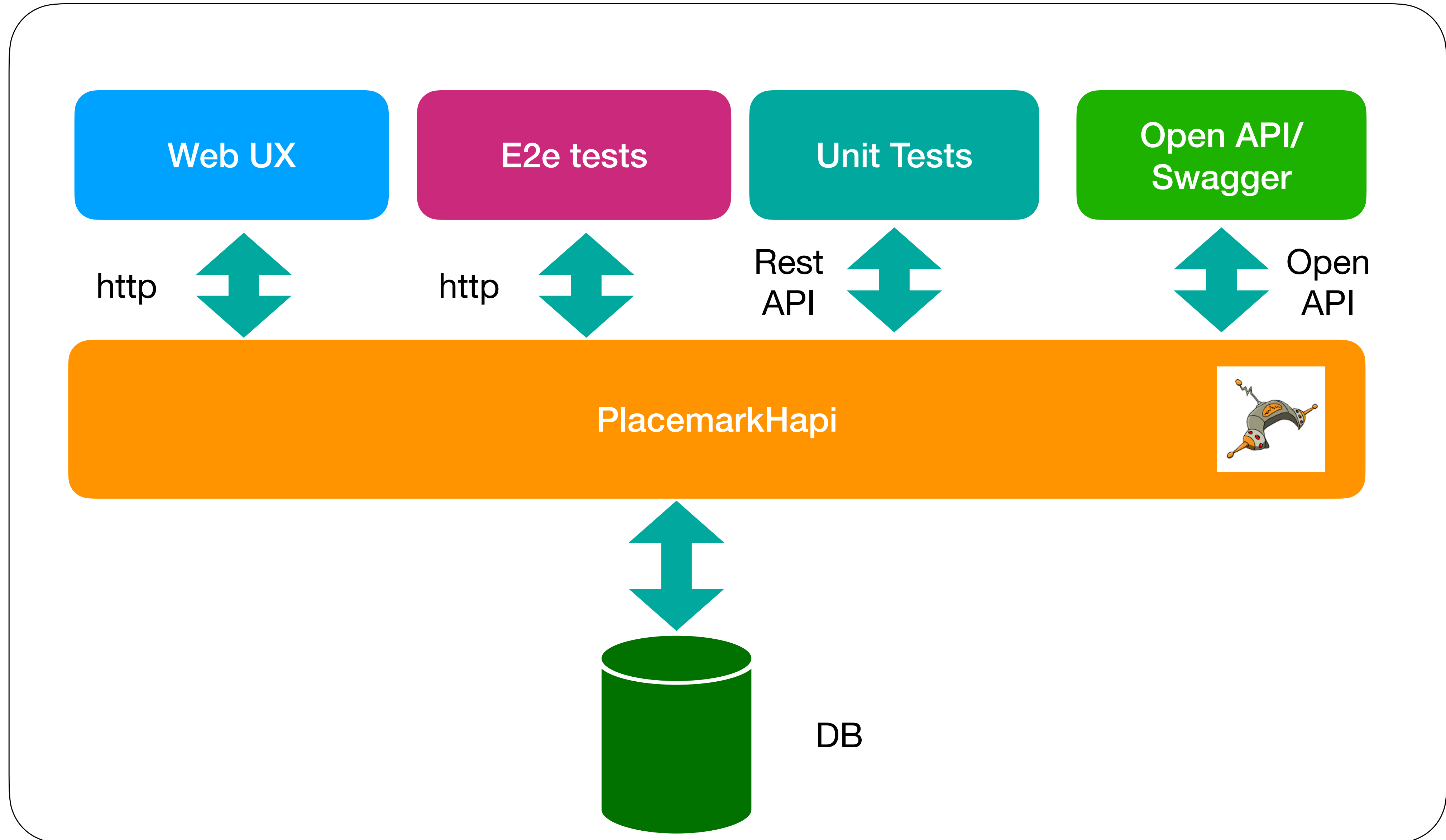


if you wish to:

- Deepen your knowledge of Hapi development
- Implement more sophisticated Models
- Pursue TDD in more depth
- Explore End-to-End testing (independently)
- Continue to evolve the features of your application



PlacemarkHAPI Architecture



PlacemarkHAPI



Grading Levels

	Features	Maps	TDD	Authentication	Devops
Level 3	Private/Public POIs	Maps with Layers (for categories)	e2e basic tests	Hashing & salting passwords	Git Flow
Level 4	Comments / Ratings	Multiple Maps per page	e2e comprehensive	OAuth (login via Github, Google etc)	Typescript
Level 5	Propose one or more innovative features, techniques or capabilities from the 5 domains (columns)				

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
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PlacemarkSveltekit



- Choose

PlacemarkSveltekit

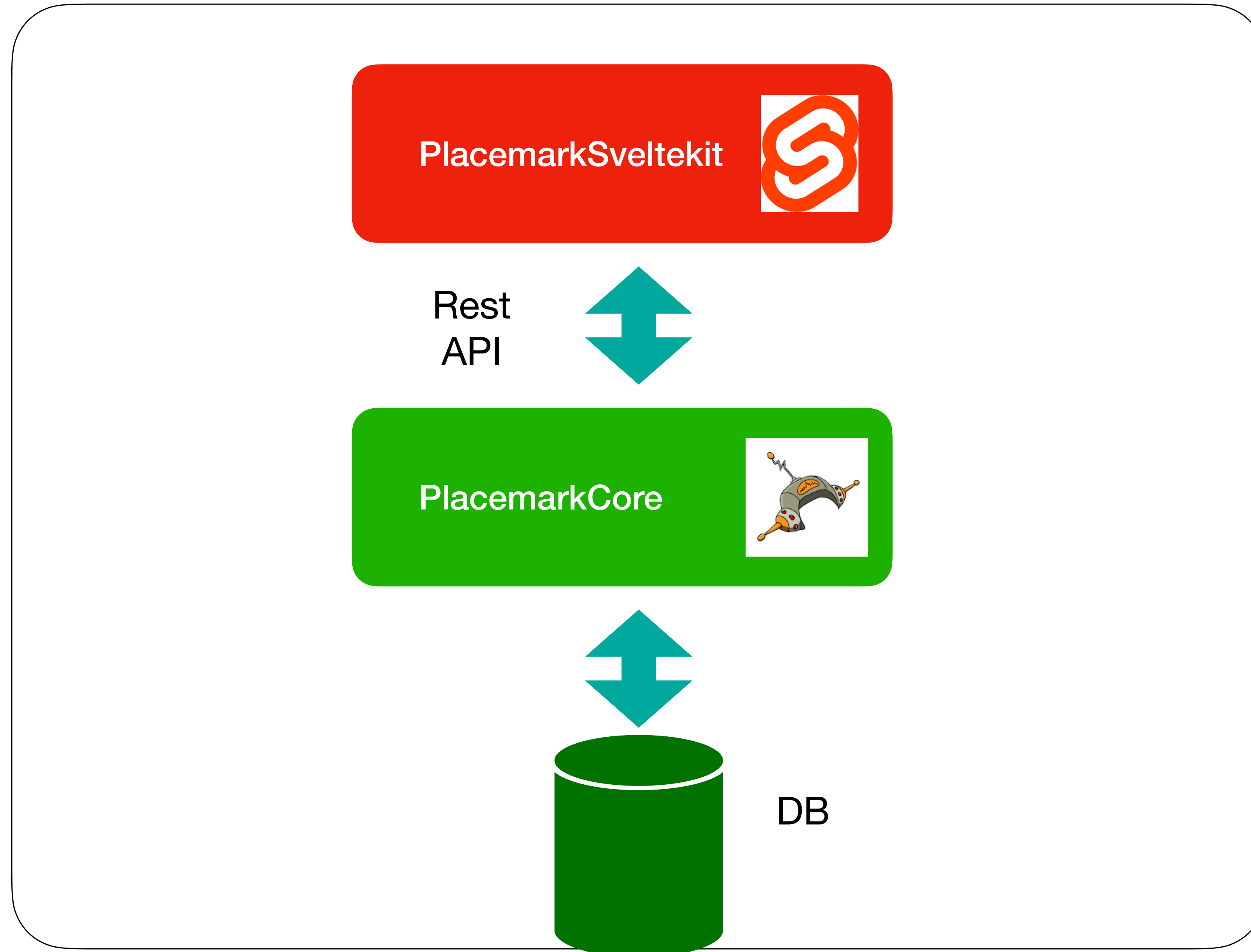


if you wish to:

- Gain an understanding of modern front end development
- Develop an engaging user experience
- Experiment with compelling Mapping features
- Explore interesting charting components
- Feature freeze your assignment 1 app (more or less)



PlacemarkApp Architecture



PlacemarkSveltekit

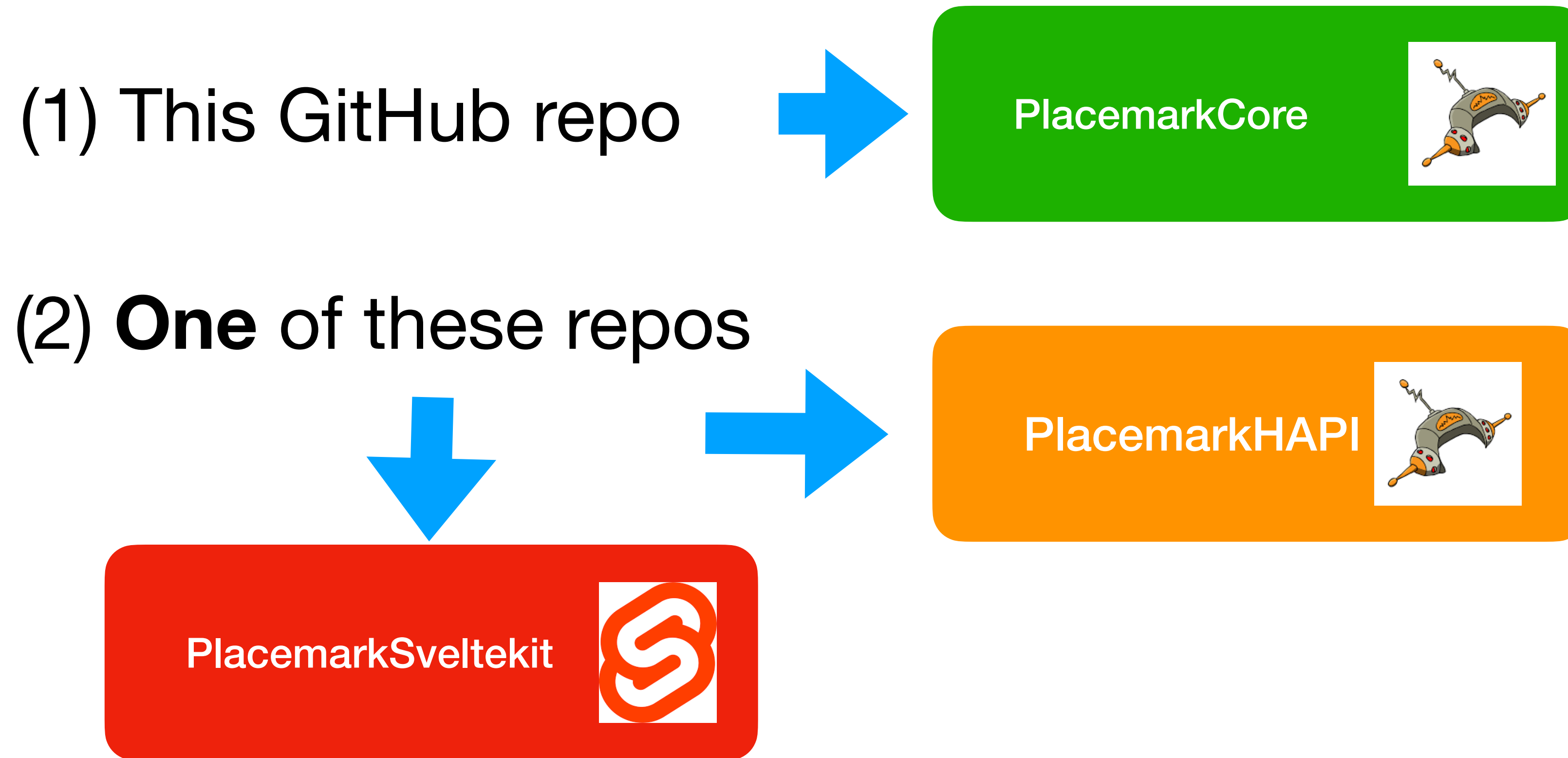


	Charts	Maps	Images	Authentication	Architecture
Level 3	Simple / Single Chart type	Maps with Layers (for categories)	Single Images per POI	Hashing & salting passwords	SvelteKit
Level 4	Multiple / Rich Chart Types	Multiple Maps per page	Multiple Images per POI	OAuth (login via Github, Google etc)	SSR
Level 5	Propose one or more innovative features, techniques or capabilities from the 5 domains (columns)				

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
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Level 5	Propose one or more innovative features, techniques or capabilities from the 5 domains (columns)				

Submissions



(3) Completed self assessment rubric (PDF doc)

(4) A video demo - 10/15 minutes (Youtube)

Level 1	30%
Level 2	55%
Level 3	65%
Level 4	75%
Level 5	100%

Approximate
Maximum Grade
for level achieved

Level 1	30%
Level 2	55%
Level 3	65%
Level 4	75%
Level 5	100%

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
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PlacemarkCore



			Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
Level 1	30%	Level 1	Signup / Login Cookie Authentication/	Name, Description, Location (lat,lng)	Basic API, Core unit Tests	JSON	Locahost	git + commit History, Structured readme
Level 2	55%	Level 2	Basic Admin Account List/Remove Users	Categories + images	Open API (Swagger) + tests	Mongo	Cloud / Render	Tagged releases, Feature Branches
Level 3	65%		Features	Maps	TDD	Authentication	Devops	
		Level 3	Private/Public POIs	Maps with Layers (for categories)	e2e basic tests	Hashing & salting passwords	Git Flow	
Level 4	75%	Level 4	Comments / Ratings	Multiple Maps per page	e2e comprehensive	OAuth (login via Github, Google etc)	Typescript	
Level 5	100%	Level 5	Propose one or more innovative features, techniques or capabilities from the 5 domains (columns)					



Level 1	30%
Level 2	55%
Level 3	65%
Level 4	75%
Level 5	100%

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
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Level 5	Propose one or more innovative features, techniques or capabilities from the 5 domains (columns)				



Assignment



Assignment Specification



Concept, detail and grading spectrum for the Assignment

Assignment Submission



Moodle assignment dropbox for the Assignment

Project Submission Template



Student Name :
Demo Video URL :
Project Repo URL :

Grade Band	Geo	Features	UX	GIT
Baseline				
Good				
Excellent				
Outstanding				

Additional Comments:

An archive of a word document to be submitted with the assignment

Advanced Full Stack Development
OTH Regensburg / SETU Ireland
Final Assignment

Student Name: _____

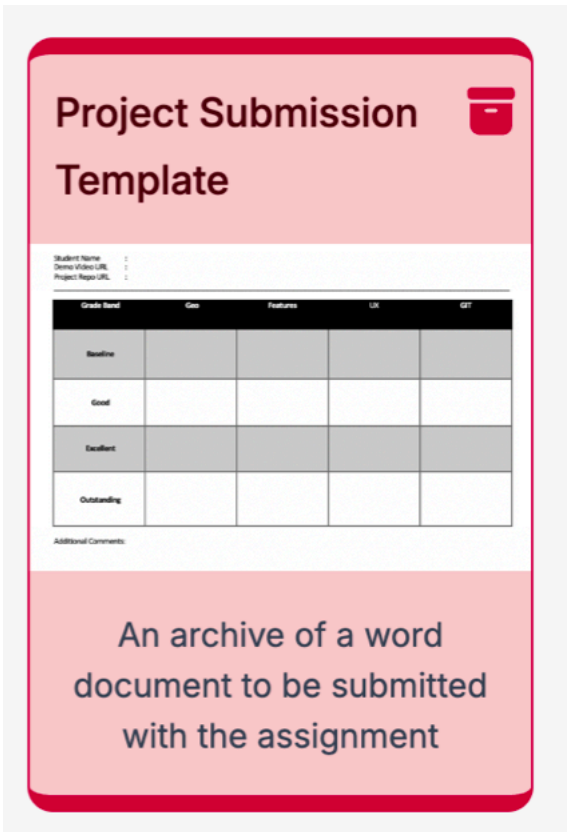
Video Demo URL: _____

PlacemarkHAPI Github URL: _____

PlacemarkSvelteKit Github URL (if attempted): _____

PlacemarkHAPI Deployed URL: _____

PlacemarkSvelteKit Deployed URL (if attempted): _____



Self Assessment Rubric

Option A or Option A

	Accounts	Placemark Features	API/Tests	Models	Deployment	DevOps
Level 1	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here
Level 2	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here
Level 3	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here
Level 4	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here	Enter your comments here
Level 5	Enter your comments here					

Project Submission Template

Student Name: _____

Student ID: _____

Project Name: _____

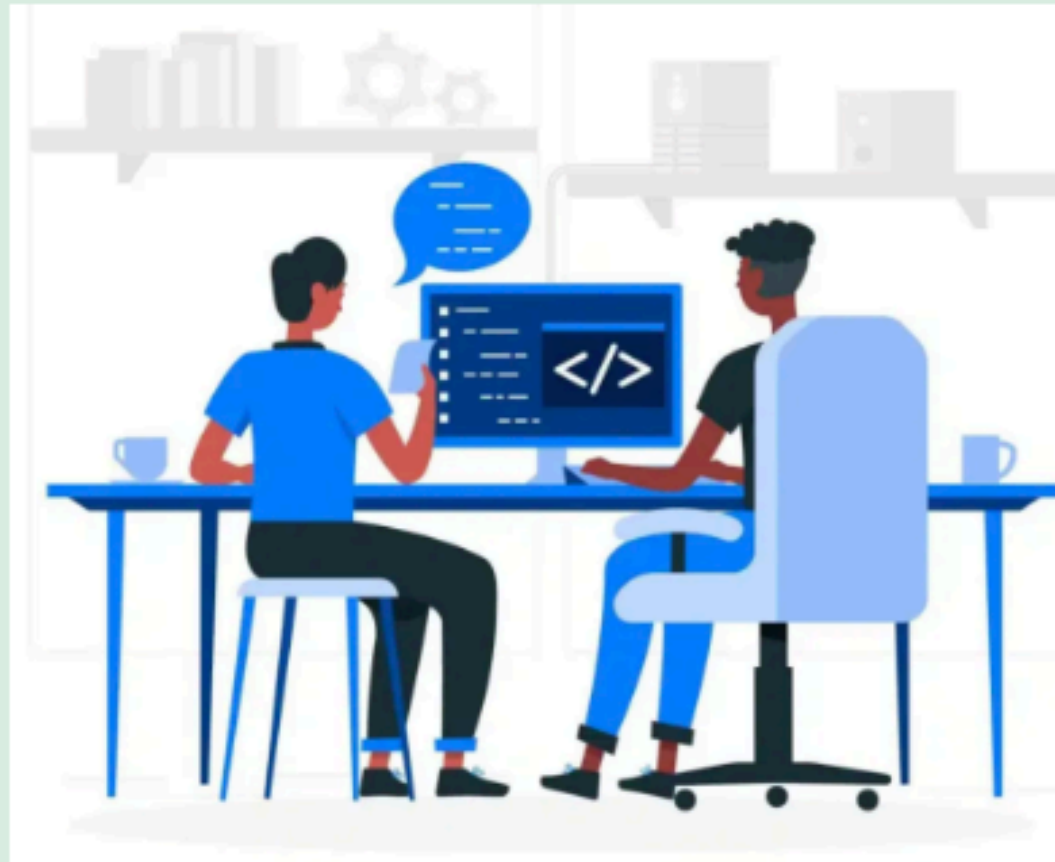
Grade Book

Week	Week	Week	Week	Week
Week				
Week				
Week				
Week				

Additional Comments

An archive of a word document to be submitted with the assignment

13: Assignment



Concept, detail and
grading rubric